

ANALYSIS HMS - FRONT OFFICE MODULE

Screen Logic Reference (01-18 Sep)

Trainer Handout | Internal logic of every FO screen & report, from analysishms-master (Laravel)

B.1 Core Data Model (6 Tables)

- guestfolio - master bill ledger; one Docid/folio_no per check-in; mFoliono links merged folios; every charge/payment hangs off its Docid.
- roomocc - actual room-stay rows; chkoutdate IS NULL = in-house; newroomno = room change; sno/sno1 handle repeat stays.
- guestprof - guest profile; reused for repeat guests (search by mobile); source of nationality/ID/GuestStatus in reports.
- paycharge - common financial ledger of FO, POS, Banquet, Laundry; VTYPE (RC=room charge, ARRES/ADRES=advance, CHK=checkout bill); AmtCr=charge, AmtDr=payment; balance = SUM(AmtCr) - SUM(AmtDr); RESTCODE tells which desk posted.
- room_mast + room_cat - rooms and categories; InclCount='Y' rooms only are counted in occupancy/inventory.
- Master tables - plan_mast (EP/CP/MAP/AP tariff), revmast (revenue heads, field_type C/P), subgroup (company/TA/OTA), busssource, gueststats, rate_list, fom_billdetail (printed bills; status='Cancel' tracks cancellations).

B.2 Master Screens - Internal Logic

- FOM Parameter: global FO settings (checkout time, RRTaxInc, folio series, night-audit lock). One wrong value affects every bill.
- Business Source: market segments; stored on guestfolio.bussource; Revenue Source Report groups revenue by this.
- Booking Source: booking.bookedby; Room Inventory joins it with bussource for market segment.
- Guest Status: guestprof.guest_status FK; gives the GuestStatus column in the Occupancy Report.
- Charge Master: revmast charge heads (Desk_code='FOM'); reports exclude special codes ROFF, DISC, CGSS/SGSS, TOUT, RMCH.
- Plan Master: EP/CP/MAP/AP; Night Audit auto-posts meal charges per planamt; Occupancy 'package' column = plan name.
- Room Features: amenity tags shown with room availability.
- Room Category: availability = NoOfRoom - occupied; rate_list is category-wise.
- Room Master: physical rooms; InclCount flag decides occupancy/inventory counting.
- Building Master: block-wise occupancy and housekeeping assignment.

B.3 Operations Screens - Internal Logic

- Blank GRC: print-only template, reads no data - same fields as Walk-in Check-in.

- Walk-in Check-in: availability from room_mast vs roomocc (chkoutdate IS NULL) -> reuse/insert guestprof -> INSERT guestfolio -> INSERT roomocc -> advance to paycharge (VTYPE=ADRES) -> room 'O'. All in one DB transaction - failure rolls back, never a half folio.
- Check-in List: guestfolio JOIN roomocc by vdate; advance = SUM(paycharge.amtcr) WHERE modeset != 'S' (settled advances excluded).
- Room Status: room_mast LEFT JOIN roomocc (chkoutdate IS NULL); rooms not found = VACANT; housekeeping tables give cleanliness.
- Bill Reprint: read-only reprint from fom_billdetail - no new voucher, no accounting effect.
- Bill Re-Settlement: old settlement reversed via refdocid trail, bill back to open, new VNO issued; entries are never deleted (audit trail).
- Misc Payment/Rect.: cash entries outside a folio (petty cash, corrections); appear in the Cashier Report.
- Merge Folio: other folios' Docids go into target's mFoliono and paycharge.FOLIONODOCID is re-pointed; Reverse Merge undoes it in order.
- In-House Room Status: live folio balance = SUM(AmtCr - AmtDr) per in-house guest.
- Add New Profile: guestprof entry without check-in; guest code reused at Walk-in/Reservation.
- Amend Stay: roomocc UPDATE; room change = old row keeps roomno, gets newroomno + chngdate, new row sno+1 - this is what makes Room Change History possible.

B.4 Reports - Query Logic (01-18 Sep data)

- Check-in Register: guestfolio.vdate range; merged folios show mFoliono; TotalGuest = adult + children; operator shown in USER column.
- Check-out Register: fom_billdetail by billdate + paycharge revenue columns via revmast (field_type='C', Desk_code='FOM').
- Cashier Report: RESTCODE='FOM'; advances appear only when not applied at check-in (DbtChkIn<>'Yes'); refdocid-linked entries skipped to avoid double count; CHK vouchers from second query; Net = AmtCr - AmtDr; mode-wise Cash/UPI/Card totals.
- Cancel Bill Details: status='Cancel' (never deleted) + reason - full fraud/audit trail.
- Occupancy Report: overlap logic - chkdate <= to-date AND (chkoutdate >= from-date OR NULL) - counts room-nights sold, not just check-ins; DayUse = same-day type='O'.
- Expected Check Out: chkoutdate IS NULL AND depdate in range; night audit's 'who leaves tomorrow' list.
- Complimentary Report: guestprof.complimentry='Y'; comp stays take zero/comp revenue code so they never double-count.
- Room Change History: roomocc self-join on newroomno chain; shows old -> new room/rate, changedby and reason.
- MIS Room Inventory: live snapshot - total stock (InclCount='Y') minus occupied = free; rack vs actual rate gap.
- MIS Sale Summary: pivot report - every FOM revenue head is a column, rows are days; ADR = RoomRent / room-nights.
- MIS Company Contribution: per company/TA per day room-nights (paycode='RMCH', vtype='RC', sno=1) - contribution matrix.
- MIS Guest Trail: one guest's full financial trail - rent, discount, extra charges, tax, outlet transfers, credit.
- MIS Company-wise Analysis: corporate room-nights, guests, gross revenue, rent, tax per company.
- MIS Revenue Source: revenue/room-nights/ADR GROUP BY bussource - OTA vs direct vs corporate mix.

- FOM Tax Detail: paycharge where paycode IN (CGSS, SGSS, luxury tax codes); GST slabs 0% / 12% (Rs. 7,500 room-rate threshold rule).

B.5 Key Code Patterns

- Permission: every screen calls revokeopen(<code>); code = MMGGSS - 14=Front Office, 11=Operations, 12=Reports, 13=MIS, 15=Tax.
- Multi-tenancy: every query filters propertyid - data isolation across hotels.
- Financial duality: AmtCr = charge (guest debit), AmtDr = payment; checkout allowed only when balance = 0.
- Date-logic gotcha: the same From-To means different columns per report - vdate (check-in), overlap (occupancy), depdate with chkoutdate IS NULL (expected checkout), billdate (cancel bill).
- Settlement chain: ADRES/ARRES advance -> DbtChkIn flag -> CHK checkout voucher -> refdocid link -> cancel = status change.
- Merge trail: mFoliono + FOLIONODOCID re-pointing + reverse-merge helper - records are never physically deleted.

B.6 Suggested 5-Day Training Flow

- Day 1 - Masters: Room Category -> Room Master -> Plan -> Charge Master -> Business Source.
- Day 2 - Operations: Room Status -> Walk-in Check-in -> In-House -> Amend Stay -> Merge Folio -> Bill Re-Settlement -> verify via Check-out Register.
- Day 3 - Reports: practice each dated report (01-18 Sep screenshots): fill filters -> read data -> explain the logic -> Excel/Print.
- Day 4 - Cash & Tax: Cashier Report reconciliation + FOM Tax Detail + GST slabs.
- Day 5 - Revision: Appendix Q&A + full live cycle (check-in to bill) on the demo property.